

1. Consider the clarity of the following proposition (Is it nice to read? What would you change about how its written while maintaining the same truth value?), then express this proposition as a sentence using symbols from propositional logic such as the quantifiers $\forall$ and $\exists$ or the connectives $\wedge, \vee, \neg$.

"If every natural number less than 50 has at most three distinct prime factors and the largest prime number smaller than 50 is 47, then the sum of the distinct primes dividing n is less than 141 for every n less than 50."

Also find the negation of this proposition both as formal logical symbols and as an English sentence.

2. Suppose that we have 4 propositions p, q, r, s for which we know the following compound propositions are true

$$p \rightarrow q, \quad q \rightarrow r, \quad r \rightarrow s, \quad \text{and} \quad s \rightarrow p.$$

Show that all of the propositions p, q, r, s are logically equivalent to each other.

3. Prove that every integer n is either a multiple of 3, one more than a multiple of 3, or 2 more than a multiple of 3. Using this, proceed to prove that for every integer n , we have that $n^4 - n^2$ is a multiple of 3.

4. Define the average of a set of k different integers $\{n_1, \dots, n_k\}$ to be their sum divided by k , i.e. the formula $(n_1 + \dots + n_k)/k$. Prove or disprove that if a collection of k integers $\{n_1, \dots, n_k\}$ has an average of N where N is not an integer, there exists an integer n_i in this collection such that $n_i < N$.